

Problem-Solution Fit

Date	2 July 2026
Team ID	
Project Name	Rising Waters – AI Based Flood Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Disaster Management Authorities - Weather Department - District Administration - Residents in Flood-Prone Areas - Emergency Response Teams	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited real-time data access - Poor internet connectivity - Limited technical resources - Delayed communication	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions: - Manual weather monitoring - Government flood alerts - Weather forecast websites Pros: - Easily available - Basic weather information Cons: - Not location-specific - Delayed alerts - Limited prediction accuracy
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Delayed flood prediction (Frequent) - Inaccurate weather analysis (Frequent) - Late emergency response (High) - Property and life loss during floods (Seasonal but Critical)	9. PROBLEM ROOT / CAUSE [RC] - Unpredictable weather changes - Delayed data processing - Lack of accurate flood prediction systems - Insufficient early warning mechanisms	7. BEHAVIOR + ITS INTENSITY [BE] - Frequently checks weather updates - Monitors rainfall during monsoon - Responds immediately to flood warnings - Shares alerts with local communities
3. TRIGGERS TO ACT [TR] - Heavy rainfall - Rising river / water levels - Weather department alerts - Extreme weather forecasts 4. EMOTIONS BEFORE / AFTER [EM] Before: - Worried - Stressed - Uncertain After: - Prepared - Safer - Confident	10. YOUR SOLUTION [SL] Rising Waters uses machine learning techniques to analyze real-time and historical data such as rainfall, temperature, humidity, and water levels to predict flood risks in advance. The system provides accurate predictions, early warning alerts, and an interactive dashboard to help authorities and communities take timely action and minimize loss of life and property.	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Rising Waters Web Application - Weather Department Website - Mobile Notifications - Social Media / Messaging Apps OFFLINE - Government Announcements - Local Disaster Management Offices - Community Volunteers - TV / Radio Broadcasts